

Q1.PROGRAM TO CALCULATE SIMPLE INTEREST.

```
C 01.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int p,r,t;
5
6      printf("Enter the principal amount: ");
7      scanf("%d",&p);
8      printf("Enter the rate of interest: ");
9      scanf("%d",&r);
10     printf("Enter the time period: ");
11     scanf("%d",&t);
12
13     printf("The simple interest is: %d \n",(p*r*t)/100);
14 }
```

Problems Output Debug Console Terminal Ports

● Enter the principal amount: 1000
Enter the rate of interest: 5
Enter the time period: 7
The simple interest is: 350
○ yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q2.PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE.

```
C 02.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int m1,m2,m3,m4,m5;
5      printf("Enter the marks of 1st subject :");
6      scanf("%d",&m1);
7
8      printf("Enter the marks of 2nd subject :");
9      scanf("%d",&m2);
10
11     printf("Enter the marks of 3rd subject :");
12     scanf("%d",&m3);
13
14     printf("Enter the marks of 4th subject :");
15     scanf("%d",&m4);
16
17     printf("Enter the marks of 5th subject :");
18     scanf("%d",&m5);
19
20
21     printf("The total marks is : %d \n", m1+m2+m3+m4+m5);
22
23     printf("The Percentage is : %d \n", (m1+m2+m3+m4+m5)/5);
24
25 }
```

Problems Output Debug Console Terminal Ports

● Enter the marks of 1st subject :46
Enter the marks of 2nd subject :68
Enter the marks of 3rd subject :89
Enter the marks of 4th subject :79
Enter the marks of 5th subject :57
The total marks is : 339
The Percentage is : 67
○ yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q3.PROGRAM TO CALCULATE GROSS SALARY.

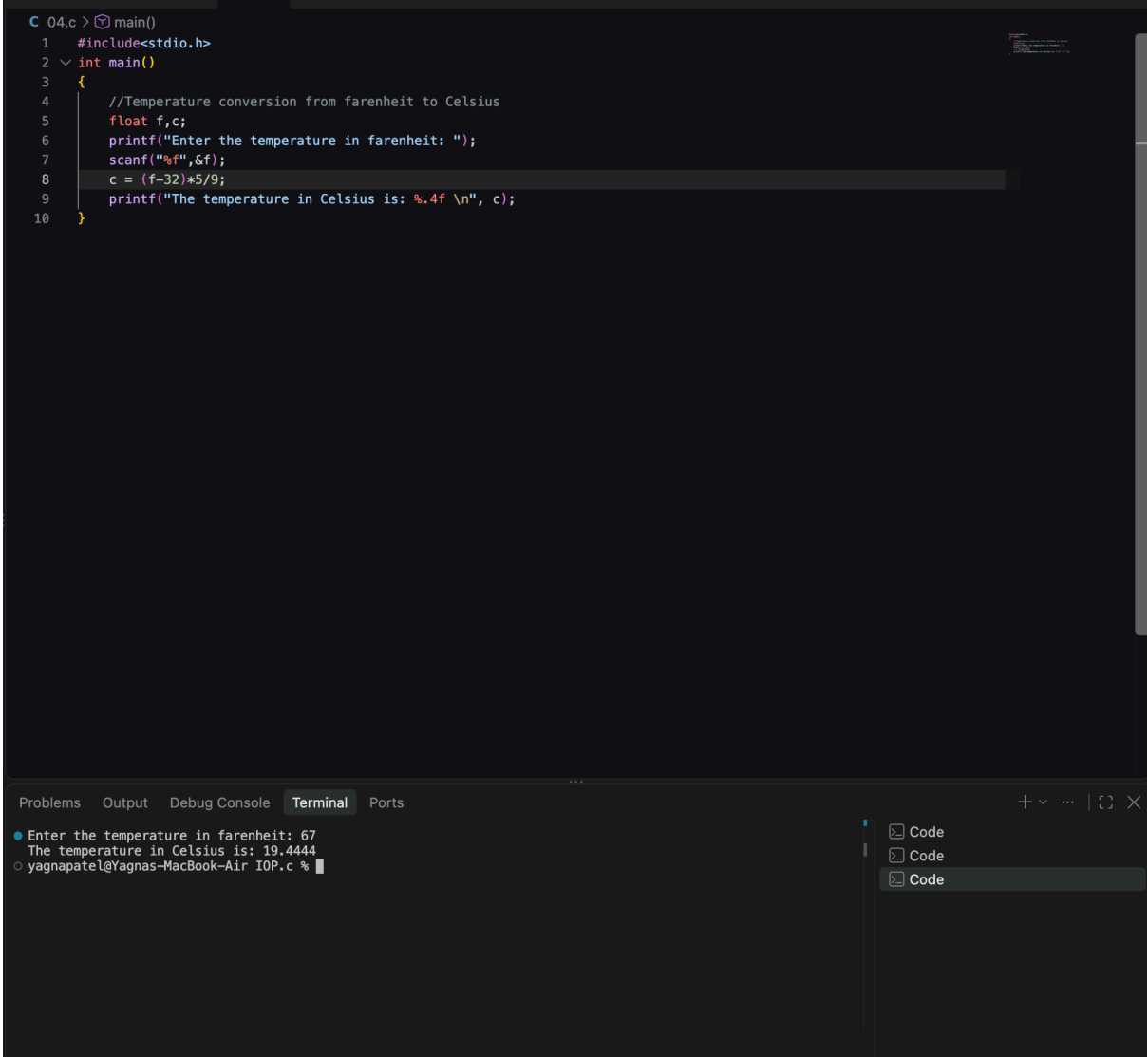
```
C 03.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      //calcuATE grossy salary
5      int BASIC, HRA, GROSS_SALARY;
6      printf("Enter the basic salary: ");
7      scanf("%d", &BASIC);
8      printf("Enter the HRA: ");
9      scanf("%d", &HRA);
10     GROSS_SALARY = BASIC + HRA;
11     printf("The gross salary is: %d \n", GROSS_SALARY);
12 }
```

Problems Output Debug Console **Terminal** Ports

● Enter the basic salary: 65000
Enter the HRA: 15000
The gross salary is: 80000
○ yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q4.PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.



The image shows a Visual Studio Code editor window with a C program for converting Fahrenheit to Celsius. The code is as follows:

```
04.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      //Temperature conversion from fahrenheit to Celsius
5      float f,c;
6      printf("Enter the temperature in fahrenheit: ");
7      scanf("%f",&f);
8      c = (f-32)*5/9;
9      printf("The temperature in Celsius is: %.4f \n", c);
10 }
```

Below the code editor, the 'Terminal' tab is active, showing the program's output:

```
Enter the temperature in fahrenheit: 67
The temperature in Celsius is: 19.4444
yagnapatel@yagnas-MacBook-Air IOP.c %
```

On the right side of the terminal, there are three 'Code' tabs, indicating that the program was run multiple times.

Q5.PROGRAM TO SWAP TOW VARIABLES USING THIRD VARIABLE.

```
C 05.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int a,b,c;
5      printf("Enter the value of a and b before swapping is :");
6      scanf("%d%d",&a,&b);
7      c=a;
8      a=b;
9      b=c;
10     printf("The value of a and b is : %d and %d\n",a,b);
11 }
```

Problems Output Debug Console **Terminal** Ports

Enter the value of a and b before swapping is :5
76
The value of a and b is : 76 and 5
yagnapatel@yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q6.PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE

```
C 06.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int a,b;
5      printf("Enter the value of a and b before swapping is :");
6      scanf("%d%d",&a,&b);
7      a = a + b;
8      b = a - b;
9      a = a - b;
10     printf("The value of a and b is : %d and %d \n",a,b);
11 }
12
```

Problems Output Debug Console Terminal Ports

```
Enter the value of a and b before swapping is :56
76
The value of a and b is : 76 and 56
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

Code
Code
Code

Q7.PROGRAM TO CALCULATE AREA OF A TRIANGLE.

```
C 07.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int base,height;
5
6      printf("Enter the value of base : ");
7      scanf("%d",&base);
8
9      printf("Enter the value of height :");
10     scanf("%d",&height);
11
12     printf("The area of the triangle is %d \n",(base*height)*1/2);
13 }
```

Problems Output Debug Console Terminal Ports

Enter the value of base : 8
Enter the value of height : 6
The area of the triangle is 24
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q8.EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS .WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY,HOURS AND MINUTES

```
08.c > [main()]
1  #include <stdio.h>
2  int main()
3  {
4      long total_sec;
5      int days, hrs, mins, sec;
6      printf("Enter total_seconds");
7      scanf("%d",&total_sec);
8
9      days = total_sec / 86400;
10     total_sec = total_sec % 86400;
11
12     hrs = total_sec / 3600;
13     total_sec = total_sec % 3600;
14
15     mins = total_sec / 60;
16     sec = total_sec % 60;
17
18     printf("Days = %d\n", days);
19     printf("Hours = %d\n", hrs);
20     printf("Minutes = %d\n", mins);
21     printf("Seconds = %d\n", sec);
22
23 }
```

Problems Output Debug Console Terminal Ports

Enter total_seconds635352
Days = 49717
Hours = 14
Minutes = 57
Seconds = 28
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q9.PROGRAM TO READ TIME IN HR,MIN,SEC AND CONVERT IT INTO TOTAL SECOND

```
C 09.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int hrs,mins,sec,total_seconds;
5      printf("Enter hrs,min,sec to convert into seconds");
6      scanf("%d %d %d",&hrs,&mins,&sec);
7
8      total_seconds = ( hrs * 3600 ) + ( mins * 60 ) + sec ;
9      printf("Total seconds is : %d",total_seconds);
10 }
```

Problems Output Debug Console **Terminal** Ports

```
● Enter hrs,min,sec to convert into seconds64
74
Total seconds is : 234314
○ yagnapatel@Yagnas-MacBook-Air IOP.c %
```

Code
Code
Code

Q10.WRIT A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA.

$$CM = M/2 + P/2 + C/2 + E$$

WHERE CM = Cut of f mark

M = Marks in Mathematics out of 200

P = Marks in Physics out of 200

C = Marks in Chemistry out of 200

E = Marks in entrance examination out of 100

```
C 10.c > main()
1  #include<stdio.h>
2  int main ()
3  {
4      int CM,M,P,C,E;
5
6      printf("Enter marks of MATHEMATICS out of 200 :");
7      scanf("%d",&M);
8
9      printf("Enter marks of PHYSICS out of 200 :");
10     scanf("%d",&P);
11
12     printf("Enter marks of CHEMISTRY out of 200 :");
13     scanf("%d",&C);
14
15     printf("Enter marks of ENTRANCE EXAMINATION out of 100 :");
16     scanf("%d",&E);
17
18     CM=(M/2) + (P/2) + (C/2) + E;
19     printf("The cutoff marks are %d\n",CM);
20 }
```

Problems Output Debug Console Terminal Ports

Enter marks of MATHEMATICS out of 200 :168
Enter marks of PHYSICS out of 200 :187
Enter marks of CHEMISTRY out of 200 :176
Enter marks of ENTRANCE EXAMINATION out of 100 :78
The cutoff marks are 343
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q11.PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

```
1  #include <stdio.h>
2  int main()
3  {
4      int Total_seconds,hrs,mins,sec;
5      printf("Enter total seconds :");
6      scanf("%d",&Total_seconds);
7
8      hrs=Total_seconds/3600;
9      mins=(Total_seconds%3600)/60;
10     sec=Total_seconds%60;
11
12     printf("The total time is %dhr %dmin %d sec\n",hrs,mins,sec);
13 }
```

Problems Output Debug Console Terminal Ports

Enter total seconds :7317381
The total time is 2032hr 36min 21 sec
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code